

# Python Basics & Strings

Pre-Assessment 1 · 13 questions · 48 points

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Score: / 48

This is a **pre-assessment**. It shows what you already know **before** we start this unit, so it is fine if some questions cover things you have not learned yet. Try every question you can. If you have no idea at all, write “**not yet**” and move on.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

## 1. Predict the output

3 points

Write exactly what this program prints. Use one answer line for each line of output.

```
print("Snack\nShop")
print('I\'m hungry')
print("Total:", 5)
```

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## 2. What type is it?

4 points

Write the type Python would report with `type()`: **int**, **float**, or **str**.

| Value | Type |
|-------|------|
| 42    |      |
| "42"  |      |
| 3.5   |      |
| 7 / 2 |      |

## 3. Good names, bad names

2 points

Circle every name that is a **legal** Python variable name. Cross out the ones that are not.

snack\_count

2cool

my-name

player1

total score

#### 4. Number crunching

5 points

Write the value of each expression.

| Expression | Value | Expression  | Value |
|------------|-------|-------------|-------|
| 17 // 5    |       | 3 + 4 * 2   |       |
| 17 % 5     |       | (3 + 4) * 2 |       |
| 2 ** 3     |       |             |       |

#### 5. f-string output

2 points

```
name = "Zara"
snacks = 3
price = 2
print(f"{name} bought {snacks} snacks for ${snacks * price}.")
```

What does the last line print?

#### 6. The birthday bug

4 points

```
age = input("How old are you? ")
next_year = age + 1
print(f"Next year you will be {next_year}!")
```

(a) The user types 11. What happens when line 2 runs? Circle one.

(1 pt)

☐ (A) It prints Next year you will be 12!

☐ (B) It prints Next year you will be 111!

☐ (C) It crashes with a TypeError

☐ (D) It prints Next year you will be 11!

(b) Explain **why** in one sentence.

(1 pt)

(c) Rewrite line 1 so the program works.

(2 pts)

## 7. Picking characters

4 points

`word = "PYTHON"`. Write what each expression gives. If it causes an error, write the error name.

| Expression             | Result |
|------------------------|--------|
| <code>len(word)</code> |        |
| <code>word[2]</code>   |        |
| <code>word[-2]</code>  |        |
| <code>word[6]</code>   |        |

## 8. Slice it

4 points

`text = "sunflower"`. Write the string each slice gives.

| Slice                   | Result |
|-------------------------|--------|
| <code>text[0:3]</code>  |        |
| <code>text[3:9]</code>  |        |
| <code>text[1:4]</code>  |        |
| <code>text[::-1]</code> |        |

## 9. Can you change a string?

2 points

```
word = "cat"
word[0] = "b"
```

(a) What happens when line 2 runs? Circle one.

(1 pt)

☐ (A) word becomes "bat"

☐ (B) Python shows an error because strings can't be changed

☐ (C) word becomes "bcat"

☐ (D) Nothing happens

(b) Write one line of code that makes `word` equal "bat" by building a new string.

(1 pt)

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## 10. Dot magic

5 points

`s = "banana split"`. Write the result of each expression.

| Expression                       | Result |
|----------------------------------|--------|
| <code>s.upper()</code>           |        |
| <code>s.title()</code>           |        |
| <code>s.count("a")</code>        |        |
| <code>s.replace("a", "o")</code> |        |
| <code>" hi ".strip()</code>      |        |

## 11. Text toolbox

5 points

`sentence = "cats like naps"`. Write the result of each expression.

| Expression                              | Result |
|-----------------------------------------|--------|
| <code>sentence.split()</code>           |        |
| <code>"-".join(["a", "b", "c"])</code>  |        |
| <code>sentence.find("like")</code>      |        |
| <code>sentence.startswith("dog")</code> |        |
| <code>"nap" in sentence</code>          |        |

## 12. Backwards Shout

3 points

Write a program that asks the user for a word, then prints the word in UPPERCASE on one line and the word **backwards** on the next line.

### EXAMPLE RUN

```
Word: robot
ROBOT
tobor
```

## 13. Tip Helper

5 points

Write a program that:

- asks for the customer's name
- asks for the price of the meal (it may have cents, like 12.50)
- works out a 20% tip (price  $\times$  0.2)
- prints a message like the one below using an **f-string**

### EXAMPLE RUN

```
Name: Ana
Meal price: 12.50
Thanks, Ana! Your tip is 2.5
```

## END OF ASSESSMENT